20260929 Quiz

for (MATH301) Modern Algebra I

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Exercise 1.7.1. Let F be a field. Show that the multiplicative group of nonzero elements of F, denoted by $F^\times$, acts on the set F by

$$g \cdot a = ga,$$

where $g \in F^\times$, $a \in F$, and ga is the usual product in F of the two field elements. State clearly which axioms in the definition of a field are used.

Solution. For $g_1, g_2 \in F^\times$ and $a \in F$, associativity of multiplication gives

$$g_1 \cdot (g_2 \cdot a) = g_1(g_2 a) = (g_1 g_2)a = (g_1 g_2) \cdot a.$$

The multiplicative identity gives

$$1 \cdot a = a.$$

Hence $F^\times$ acts on F. The action axioms use associativity of multiplication and the multiplicative identity. The field axioms also ensure that the nonzero elements are closed under multiplication and inverses, so $F^\times$ is a group.

□

Exercise 1.7.16. Let G be any group and let $A = G$. Show that the maps defined by

$$g \cdot a = gag^{-1}$$

for all $g, a \in G$ satisfy the axioms of a left group action. This action of G on itself is called *conjugation*.

Solution. For $g_1, g_2, a \in G$,

$$\begin{aligned} g_1 \cdot (g_2 \cdot a) &= g_1(g_2 a g_2^{-1})g_1^{-1} \\ &= (g_1 g_2)a(g_1 g_2)^{-1} \\ &= (g_1 g_2) \cdot a. \end{aligned}$$

Also

$$1 \cdot a = 1a1^{-1} = a.$$

Thus conjugation defines a left group action of G on itself.

□

Exercise 1.7.17. Let G be a group and let G act on itself by left conjugation, so each $g \in G$ maps G to G by

$$x \mapsto gxg^{-1}.$$

For fixed $g \in G$, prove that conjugation by g is an isomorphism from G onto itself, i.e., an automorphism of G (cf. Exercise 1.6.20). Deduce that x and gxg^{-1} have the same order for all x in G and that for any subset A of G,

$$|A| = |gAg^{-1}|,$$

where $gAg^{-1} = \{gag^{-1} \mid a \in A\}$.

Solution. For fixed $g \in G$, define $\varphi_g(x) = gxg^{-1}$. Then

$$\varphi_g(xy) = gxyg^{-1} = (gxg^{-1})(gyg^{-1}) = \varphi_g(x)\varphi_g(y),$$

so φ_g is a homomorphism. Conjugation by g^{-1} is its inverse, since

$$\varphi_{g^{-1}}(\varphi_g(x)) = x.$$

Thus φ_g is an automorphism. Isomorphisms preserve element orders, so

$$|x| = |gxg^{-1}|.$$

Moreover, the restriction of φ_g gives a bijection from A onto gAg^{-1}, hence

$$|A| = |gAg^{-1}|.$$

□

Exercise 2.1.2. In each of (a)–(e), prove that the specified subset is not a subgroup of the given group.

(a) The set of 2-cycles in S_n for $n \geq 3$.

(b) The set of reflections in D_n for $n \geq 3$.

(c) For a composite integer $n > 1$ and a group G containing an element of order n, the set

$$\{x \in G \mid |x| = n\} \cup \{1\}.$$

(d) The set of positive and negative odd integers in $\mathbb{Z}$, together with 0, under addition.

(e) The set of real numbers whose square is a rational number, under addition.

Notation. Following MATH000, D_n denotes the dihedral group of symmetries of a regular n-gon, so $|D_n| = 2n$. Dummit–Foote denotes the same group by D_{2n}.

Solution.

(a) The identity permutation is not a 2-cycle, so the set does not contain the identity.

(b) The identity is not a reflection, so the set does not contain the identity.

(c) Let $x \in G$ have order n. Since n is composite, write $n = st$ with $1 < s, t < n$. Then Exercise 1.1.23 gives

$$|x^s| = t.$$

Thus x belongs to the set but x^s does not. A subgroup containing x must contain all powers of x, so the set is not a subgroup.

(d) The element 1 belongs to the set, but

$$1 + 1 = 2$$

does not. Hence the set is not closed under addition.

(e) Both 1 and $\sqrt{2}$ belong to the set, but

$$(1 + \sqrt{2})^2 = 3 + 2\sqrt{2} \notin \mathbb{Q}.$$

Hence $1 + \sqrt{2}$ does not belong to the set, so it is not closed under addition.

□

Exercise 2.1.3. Show that the following subsets of the dihedral group D_4 are subgroups:

$$\{1, r^2, s, sr^2\}, \qquad \{1, r^2, sr, sr^3\}.$$

Solution. Let

$$H = \{r^{2i}, sr^{2i} \mid i = 0, 1\}.$$

For $i, j \in \{0, 1\}$, using $r^4 = s^2 = 1$ and $r^i s = sr^{-i}$,

$$\begin{aligned} r^{2i}r^{2j} &= r^{2(i+j)}, & r^{2i}(sr^{2j}) &= sr^{2(j-i)}, \\ (sr^{2i})r^{2j} &= sr^{2(i+j)}, & (sr^{2i})(sr^{2j}) &= r^{2(j-i)}. \end{aligned}$$

Thus H is closed under multiplication. Since H is finite and nonempty, $H \leq D_4$.

Similarly, let

$$K = \{r^{2i}, sr^{2i+1} \mid i = 0, 1\}.$$

Then

$$\begin{aligned} r^{2i}r^{2j} &= r^{2(i+j)}, & r^{2i}(sr^{2j+1}) &= sr^{2(j-i)+1}, \\ (sr^{2i+1})r^{2j} &= sr^{2(i+j)+1}, & (sr^{2i+1})(sr^{2j+1}) &= r^{2(j-i)}. \end{aligned}$$

Hence K is closed under multiplication, and since K is finite and nonempty, $K \leq D_4$.

□

Exercise 2.1.6. Let G be an abelian group. Prove that

$$\{g \in G \mid |g| < \infty\}$$

is a subgroup of G, called the *torsion subgroup* of G. Give an explicit example where this set is not a subgroup when G is nonabelian.

Solution. Let

$$H = \{g \in G \mid |g| < \infty\}.$$

Clearly $1 \in H$. If $x, y \in H$ with $|x| = m$ and $|y| = n$, then, since G is abelian,

$$(xy^{-1})^{mn} = x^{mn}y^{-mn} = 1.$$

Thus $xy^{-1} \in H$, so the Subgroup Criterion gives $H \leq G$.

For a nonabelian example, in $\mathrm{GL}_2(\mathbb{R})$ let

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 0 & 2 \\ \frac{1}{2} & 0 \end{pmatrix}.$$

Then $\mathbf{A}^2 = \mathbf{B}^2 = \mathbf{I}$, but

$$\mathbf{AB} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{pmatrix}, \qquad (\mathbf{AB})^k = \begin{pmatrix} 2^{-k} & 0 \\ 0 & 2^k \end{pmatrix} \neq \mathbf{I}$$

for every $k > 0$, so $\mathbf{AB}$ has infinite order. Hence the finite-order elements need not form a subgroup in a nonabelian group.

□

Exercise 2.1.13. Let H be a subgroup of the additive group of rational numbers with the property that $1/x \in H$ for every nonzero element x of H. Prove that

$$H = \{0\} \quad \text{or} \quad H = \mathbb{Q}.$$

Solution. Assume $H \neq \{0\}$ and choose

$$0 \neq x = \frac{a}{b} \in H,$$

where $b > 0$ and $\gcd(|a|, b) = 1$. By hypothesis, $1/x = b/a \in H$. Since H is an additive subgroup,

$$bx = a \in H, \qquad a\left(\frac{1}{x}\right) = b \in H.$$

By Bézout's identity there exist $m, n \in \mathbb{Z}$ such that $ma + nb = 1$, so $1 \in H$. Hence every integer belongs to H. For any nonzero $q \in \mathbb{Z}$, the hypothesis then gives $1/q \in H$, and therefore

$$\frac{p}{q} = p\left(\frac{1}{q}\right) \in H$$

for every $p/q \in \mathbb{Q}$. Thus $H = \mathbb{Q}$.

□

Exercise 2.3.1. Find all subgroups of $C_{45} = \langle x \rangle$, giving a generator for each. Describe the containments between these subgroups.

Notation. Following MATH000, C_n denotes a cyclic group of order n. Dummit–Foote denotes the same abstract cyclic group by Z_n. The notation $\mathbb{Z}/n\mathbb{Z}$ denotes the residue classes modulo n; when used here as a group, it is understood with addition. Residue classes are denoted by $[a]_n$.

Solution. The positive divisors of 45 are

$$1, 3, 5, 9, 15, 45.$$

A cyclic group has exactly one subgroup of order d for each divisor d of its order, namely $\langle x^{45/d} \rangle$. Thus the subgroups are

order	subgroup
1	$\{1\} = \langle x^{45} \rangle$
3	$\langle x^{15} \rangle$
5	$\langle x^{9} \rangle$
9	$\langle x^{5} \rangle$
15	$\langle x^{3} \rangle$
45	$\langle x \rangle$

If H_d denotes the subgroup of order d, then

$$H_d \leq H_e \quad \Longleftrightarrow \quad d \mid e.$$

This completely describes the containments.

□

Exercise 2.3.8. Let $C_{48} = \langle x \rangle$. For which integers a does the map φ_a defined by

$$\varphi_a : [1]_{48} \mapsto x^a$$

extend to an isomorphism from $\mathbb{Z}/48\mathbb{Z}$ onto C_{48}?

Solution. For every integer a, the relation $48[1]_{48} = [0]_{48}$ is respected because $(x^a)^{48} = 1$, so $[1]_{48} \mapsto x^a$ extends to a homomorphism. It is an isomorphism exactly when x^a is a generator of C_{48}. For a cyclic group of order 48,

$$|x^a| = \frac{48}{\gcd(a, 48)}.$$

Hence

$$\varphi_a \text{ is an isomorphism} \quad \Longleftrightarrow \quad \gcd(a, 48) = 1.$$

Equivalently,

$$a \equiv 1, 5, 7, 11, 13, 17, 19, 23, 25, 29, 31, 35, 37, 41, 43, 47 \pmod{48}.$$

□

Exercise 2.3.11. Find all cyclic subgroups of D_4. Find a proper subgroup of D_4 which is not cyclic.

Solution. The elements of D_4 are

$$1, r, r^2, r^3, s, sr, sr^2, sr^3,$$

where $|r| = 4$, $|r^2| = 2$, and every reflection has order 2. Hence the distinct cyclic subgroups are

$$\{1\}, \qquad \langle r \rangle, \qquad \langle r^2 \rangle, \qquad \langle s \rangle, \qquad \langle sr \rangle, \qquad \langle sr^2 \rangle, \qquad \langle sr^3 \rangle.$$

By Exercise 2.1.3,

$$H = \{1, r^2, s, sr^2\}$$

is a proper subgroup of D_4. Every nonidentity element of H has order 2, so no element generates H. Thus H is not cyclic.

□

Exercise 2.3.12. Prove that the following groups are not cyclic:

(a) $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$,

(b) $(\mathbb{Z}/2\mathbb{Z}) \times \mathbb{Z}$,

(c) $\mathbb{Z} \times \mathbb{Z}$.

Solution.

(a) Every nonidentity element of $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$ has order 2. A cyclic group of order 4 would have an element of order 4, so this group is not cyclic.

(b) Suppose $(\mathbb{Z}/2\mathbb{Z}) \times \mathbb{Z} = \langle (a, m) \rangle$. Since $(0, 1)$ must be generated, there is $k \in \mathbb{Z}$ such that

$$k(a, m) = (0, 1),$$

so $km = 1$. Hence $m = \pm 1$. But then the only multiple of (a, m) with second coordinate 0 is the zero multiple, so $(1, 0)$ cannot be generated. This is a contradiction.

(c) Suppose $\mathbb{Z} \times \mathbb{Z} = \langle (a, b) \rangle$. Since $(1, 0)$ must be generated, there is $k \in \mathbb{Z}$ such that

$$k(a, b) = (1, 0).$$

Thus $ka = 1$ and $kb = 0$, so $k = \pm 1$ and $b = 0$. Then every multiple of (a, b) has second coordinate 0, so $(0, 1)$ cannot be generated. Hence $\mathbb{Z} \times \mathbb{Z}$ is not cyclic.

□

Exercise 2.3.19. Show that if H is any group and h is an element of H, then there is a unique homomorphism from $\mathbb{Z}$ to H such that

$$1 \mapsto h.$$

Solution. Define

$$\varphi \colon \mathbb{Z} \to H, \qquad \varphi(n) = h^n.$$

Then

$$\varphi(m + n) = h^{m+n} = h^m h^n = \varphi(m)\varphi(n),$$

so φ is a homomorphism and $\varphi(1) = h$.

If $\psi \colon \mathbb{Z} \to H$ is any homomorphism with $\psi(1) = h$, then for every $n \in \mathbb{Z}$,

$$\psi(n) = \psi(1)^n = h^n = \varphi(n).$$

Thus φ is unique.

□

Exercise 2.3.21. Let p be an odd prime and let n be a positive integer. Use the Binomial Theorem to show that

$$(1 + p)^{p^{n-1}} \equiv 1 \pmod{p^n}$$

but

$$(1 + p)^{p^{n-2}} \not\equiv 1 \pmod{p^n}.$$

Deduce that $1 + p$ is an element of order p^{n-1} in the multiplicative group $(\mathbb{Z}/p^n\mathbb{Z})^\times$.

Clarification. The second congruence is understood for $n \geq 2$. For $n = 1$, the final order statement is immediate.

Solution. For $m \geq 0$, we claim that

$$(1 + p)^{p^m} \equiv 1 + p^{m+1} \pmod{p^{m+2}}.$$

The case $m = 0$ is immediate. Suppose it holds for m, so

$$(1 + p)^{p^m} = 1 + p^{m+1} + cp^{m+2} = 1 + u$$

for some $c \in \mathbb{Z}$. By the Binomial Theorem,

$$(1 + u)^p = 1 + pu + \sum_{j=2}^{p-1} \binom{p}{j} u^j + u^p.$$

Since p is prime, $p \mid \binom{p}{j}$ for $1 \leq j \leq p - 1$, and all terms after pu are divisible by p^{m+3}. Hence

$$(1 + p)^{p^{m+1}} \equiv 1 + p^{m+2} \pmod{p^{m+3}},$$

proving the claim.

Taking $m = n - 1$ gives

$$(1+p)^{p^{n-1}} \equiv 1 \pmod{p^n}.$$

For $n \geq 2$, taking $m = n - 2$ gives

$$(1+p)^{p^{n-2}} \equiv 1 + p^{n-1} \not\equiv 1 \pmod{p^n}.$$

Thus, for $n \geq 2$, the order of $1 + p$ divides p^{n-1} but does not divide p^{n-2}, so it is p^{n-1}. For $n = 1$, $1 + p \equiv 1 \pmod{p}$, whose order is $1 = p^0$.

□

Exercise 2.3.24. Let G be a finite group and let $x \in G$.

(a) Prove that if $g \in N_G(\langle x \rangle)$, then

$$gxg^{-1} = x^a$$

for some $a \in \mathbb{Z}$.

(b) Prove conversely that if $gxg^{-1} = x^a$ for some $a \in \mathbb{Z}$, then $g \in N_G(\langle x \rangle)$.
Hint. First show that $gx^k g^{-1} = (gxg^{-1})^k = x^{ak}$ for any integer k, so that $g\langle x \rangle g^{-1} \leq \langle x \rangle$. If x has order n, show that the elements $gx^i g^{-1}$, $i = 0, 1, \ldots, n-1$, are distinct, so that

$$|g\langle x \rangle g^{-1}| = |\langle x \rangle| = n,$$

and conclude that $g\langle x \rangle g^{-1} = \langle x \rangle$.

Solution.

(a) If $g \in N_G(\langle x \rangle)$, then

$$g\langle x \rangle g^{-1} = \langle x \rangle.$$

Since $x \in \langle x \rangle$, it follows that $gxg^{-1} \in \langle x \rangle$, so

$$gxg^{-1} = x^a$$

for some $a \in \mathbb{Z}$.

(b) Suppose $gxg^{-1} = x^a$. For every $k \in \mathbb{Z}$,

$$gx^k g^{-1} = (gxg^{-1})^k = x^{ak} \in \langle x \rangle,$$

hence

$$g\langle x \rangle g^{-1} \leq \langle x \rangle.$$

Let $|x| = n$. If

$$gx^i g^{-1} = gx^j g^{-1}$$

for $0 \leq i, j < n$, cancellation gives $x^i = x^j$, hence $i = j$. Thus

$$|g\langle x \rangle g^{-1}| = n = |\langle x \rangle|.$$

The subgroup $g\langle x \rangle g^{-1}$ is contained in $\langle x \rangle$ and has the same finite order, so the inclusion is an equality. Hence

$$g\langle x \rangle g^{-1} = \langle x \rangle,$$

so $g \in N_G(\langle x \rangle)$.

□

Exercise 2.3.26. Let C_n be a cyclic group of order n, and for each integer a let

$$\sigma_a \colon C_n \to C_n, \qquad \sigma_a(x) = x^a$$

for all $x \in C_n$.

(a) Prove that σ_a is an automorphism of C_n if and only if a and n are relatively prime (cf. Exercise 1.6.20).

(b) Prove that $\sigma_a = \sigma_b$ if and only if $a \equiv b \pmod{n}$.

(c) Prove that every automorphism of C_n is equal to σ_a for some integer a.

(d) Prove that $\sigma_a \circ \sigma_b = \sigma_{ab}$. Deduce that the map

$$[a]_n \mapsto \sigma_a$$

is an isomorphism of $(\mathbb{Z}/n\mathbb{Z})^\times$ onto $\mathrm{Aut}(C_n)$. Thus $\mathrm{Aut}(C_n)$ is an abelian group of order $\varphi(n)$.

Solution. Let $C_n = \langle z \rangle$.

(a) Since C_n is abelian,

$$\sigma_a(xy) = (xy)^a = x^a y^a = \sigma_a(x)\sigma_a(y),$$

so σ_a is a homomorphism. Its image is $\langle z^a \rangle$. Hence σ_a is an automorphism if and only if z^a generates C_n, which holds if and only if

$$\gcd(a, n) = 1.$$

(b) If $\sigma_a = \sigma_b$, then evaluating at z gives $z^a = z^b$, so

$$z^{a-b} = 1.$$

Since $|z| = n$, this is equivalent to $n \mid (a - b)$, i.e., $a \equiv b \pmod{n}$. The converse follows immediately from $x^n = 1$ for every $x \in C_n$.

(c) Let $\sigma \in \mathrm{Aut}(C_n)$. Since z has order n and automorphisms preserve element orders, $\sigma(z)$ is a generator of C_n. Thus

$$\sigma(z) = z^a$$

for some integer a. For every $k \in \mathbb{Z}$,

$$\sigma(z^k) = \sigma(z)^k = z^{ak} = \sigma_a(z^k),$$

so $\sigma = \sigma_a$.

(d) For every $x \in C_n$,

$$(\sigma_a \circ \sigma_b)(x) = \sigma_a(x^b) = x^{ab} = \sigma_{ab}(x).$$

Hence $\sigma_a \circ \sigma_b = \sigma_{ab}$. By parts (a)–(c), the map

$$(\mathbb{Z}/n\mathbb{Z})^\times \to \mathrm{Aut}(C_n), \qquad [a]_n \mapsto \sigma_a,$$

is well defined, bijective, and a homomorphism. Therefore

$$(\mathbb{Z}/n\mathbb{Z})^\times \cong \mathrm{Aut}(C_n).$$

Consequently $\mathrm{Aut}(C_n)$ is abelian and has order $\varphi(n)$.

□